2.6 Investigating the Representation of Socially Undesirable Personality Traits

While the previous sections focused on general patterns within the personality vector space, this section addresses a critical ethical consideration: the representation of socially undesirable or potentially harmful personality traits. Specifically, we focus on traits grouped within Cluster 8 from our previous clustering analysis (e.g., "lustful," "masochistic," "pedophilic"), which raise concerns about potential biases and ethical implications.

2.6.1 Visualizing the Subspace of Undesirable Traits

To gain a deeper understanding of how these sensitive traits are positioned within the broader personality space, we employed dimensionality reduction techniques for visualization. We selected the traits belonging to *Cluster 8* and projected them onto a three-dimensional space using PCA. This visualization allows us to examine the relative positioning, clustering, and potential relationships among these traits within a more interpretable representation.

2.6.2 Identifying Proximal Personality Traits

Beyond simply visualizing the subspace of socially undesirable traits, a crucial question arises: which other personality attributes reside in close proximity to this region of the activation space? Understanding the traits that border this "undesirable" cluster could provide valuable insights into the potential pathways or precursors to more harmful characteristics.

To address this, we calculated the distance of all personality traits in our lexicon to the centroid of Cluster 8 within the dimensionally reduced space. Table 4 presents the top five traits consistently found to be closest to the undesirable cluster across multiple dimensionality reduction techniques (PCA and t-SNE).

Proximity	PCA	t-SNE	UMAP
1	competitive	vindictive	neurotic
2	confident	machiavellian	machiavellian
3	determined	independent	manipulative
4	bold	manipulative	vindictive
5	tenacious	self-centered	schizotypal

Table 4: Top five personality traits consistently ranked closest to the centroid of Cluster 8 (undesirable traits) across multiple dimensionality reduction techniques.

Figure 9 provides a visual representation of these proximal traits using both PCA and t-SNE. The results reveal a pattern of personality characteristics that, while not inherently negative, often manifest in ways that can be perceived as manipulative, assertive, or self-serving. This proximity

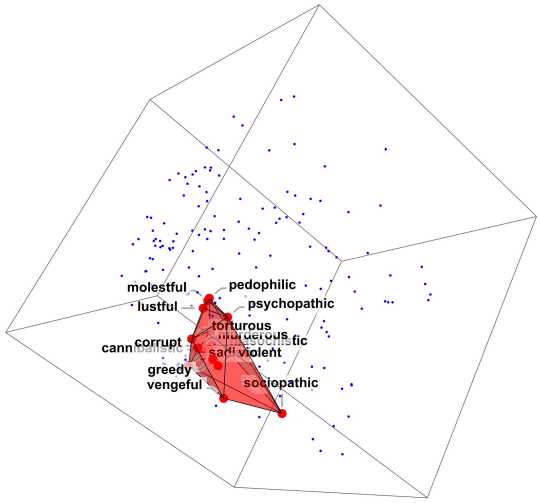

Figure 8: Three-dimensional visualization of the personality traits belonging to Cluster 8 (associated with socially undesirable characteristics) after dimensionality reduction using PCA.

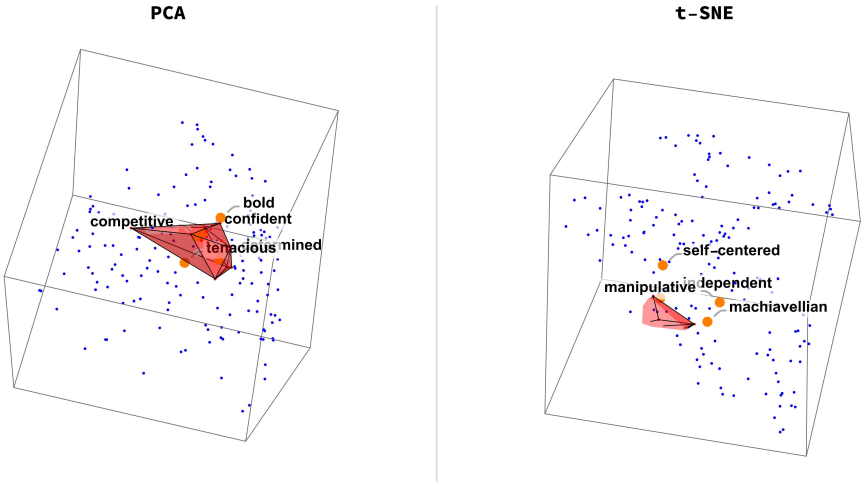

Figure 9: Visualization of the personality traits closest to Cluster 8 in a two-dimensional space using both PCA and t-SNE. Traits closer to the center (Cluster 8) are more proximal in the original activation space.

highlights the potential for these traits to be associated with or even contribute to the emergence of more socially undesirable behaviors.

2.6.3 Ethical Implications and Future Directions

These findings have significant implications for the development and deployment of LLMs. By understanding the traits that border the "undesirable" region of the personality space, we gain insights into potential precursors to harmful content generation. This knowledge is crucial for:

- **Bias Mitigation:** Developing techniques to disentangle these proximal traits from the undesirable cluster could help reduce the likelihood of the model associating neutral or positive attributes with harmful ones.
- **Safety Mechanisms:** Incorporating this knowledge into safety mechanisms could enable the detection of potentially problematic language patterns before they escalate into more severe forms.
- **Responsible AI Design:** This understanding underscores the importance of carefully considering personality representations in LLM development, advocating for approaches that promote ethical and socially responsible language generation.

Further research is needed to explore the nuances of these relationships, develop effective mitigation strategies, and ensure that LLMs are designed and utilized in a manner that minimizes potential harm while maximizing their potential for positive societal impact.

3 Interactive Personality Control

To demonstrate the practical application of our personality manipulation methodology, we developed an interactive chat user interface (UI). This interface allows users to explore the effects of different personality traits on the LLM's conversational style in real-time.

3.1 Exploring Base Personalities

We first created a chat interface that allows users to interact with the LLM using pre-defined base personalities identified in our earlier analysis. This interface enables conversations where the LLM's responses are imbued with a selected personality trait, such as "charismatic," "humble," or "outgoing." Figure 10 shows examples of this interface with different personality traits.

3.2 Modular Personality Design

Building upon the base personality interactions, we developed a more advanced interface leveraging insights from our principal component analysis. This interface provides users with fine-grained control over the LLM's personality, allowing them to create custom personality profiles by adjusting